

From Room-Sized Machines to Your Pocket

The extraordinary story of how computers developed — and why they changed everything

semiconductor • transistor • microprocessor • mainframe • integrated circuit • binary code • operating system • artificial intelligence • processor • algorithm • data storage • programming language • hardware • software

The device you are using right now — whether a smartphone, laptop, or tablet — is one of the most sophisticated objects ever created by human beings. It contains billions of tiny components, performs millions of calculations every second, and connects you instantly to almost all of human knowledge. Yet seventy years ago, a computer was a machine the size of a large room, weighed several tonnes, and could perform only the most basic calculations. The story of how we got from there to here is one of the most remarkable stories of human **innovation** in history.

Computers did not simply appear fully-formed. They were the result of decades of scientific breakthroughs, engineering challenges, and visionary thinking — each generation of machines making the next one possible. Understanding this history helps us appreciate not just where computers came from, but where they are taking us next. Before we explore the full story, consider these two questions:

1 You are asked to imagine a computer from 1950. What do you picture?

- a Something similar to today's laptop — small, fast, and easy to use.
- b An enormous machine filling an entire room, operated by a team of engineers, used only by governments or universities.

2 Why do you think computers became smaller and more powerful over time?

- a Companies simply made them smaller to be more convenient — the technology was always possible.
- b Each scientific breakthrough — from the transistor to the microprocessor — made smaller, faster computers possible for the first time.

If you chose b) both times, you are already thinking like a historian of technology. The development of computers was not a single invention — it was a chain of discoveries, each one unlocking the next. Let us follow that chain from the very beginning.

The first true electronic computers were built during and just after the Second World War. They were created to solve specific military and scientific problems — calculating artillery trajectories, breaking enemy codes, and processing complex mathematical equations far faster than any human could. The most famous of these early machines was ENIAC, built in the United States in 1945. It weighed around 27 tonnes, contained nearly 18,000 vacuum tubes, and filled a room the size of a large apartment. Yet by modern standards it was incredibly slow — a basic smartphone today is billions of times more powerful.

These early computers stored and processed data using **binary code** — a system of just two values, 0 and 1, that represents all information as a series of electrical signals. This fundamental principle has not changed to this day. Every image, every word, every song stored on your phone is ultimately represented as a sequence of billions of 0s and 1s. What has changed almost beyond recognition is the speed and miniaturisation of the **hardware** that processes them.

"ENIAC filled an entire room, weighed 27 tonnes, and needed a team of engineers to operate it. A modern smartphone performs the same calculations in a fraction of a second — and fits in your pocket."

TIMELINE • KEY MOMENTS IN COMPUTER HISTORY

1945	ENIAC — the first general-purpose electronic computer Built at the University of Pennsylvania, USA. Weighed 27 tonnes. Used vacuum tubes to process data.
1947	The transistor is invented Bell Labs scientists develop the transistor — a tiny switch that controls electrical signals, far smaller and more reliable than vacuum tubes.
1958	The integrated circuit is created Jack Kilby at Texas Instruments combines multiple transistors on a single chip of semiconductor material, making electronics dramatically smaller.
1971	The first microprocessor Intel releases the 4004 — the first commercial microprocessor, placing an entire computer processor on a single chip. This makes personal computers possible.
1981	The IBM Personal Computer IBM launches the first widely-adopted personal computer, running an operating system called MS-DOS. The era of the home computer begins.
1991	The World Wide Web goes public British scientist Tim Berners-Lee makes the World Wide Web available to the public, transforming computers from standalone machines into a global network.

2007	The first iPhone Apple launches the iPhone — combining a phone, computer, and internet device in one pocket-sized package. The smartphone era begins.
2022	The rise of artificial intelligence Large language models and AI tools become widely available to the public, beginning a new chapter in the story of what computers can do.

PART TWO · THE TRANSISTOR REVOLUTION (1950s–1970s)

The single most important development in the history of computing was almost certainly the invention of the **transistor** in 1947. A transistor is a tiny electronic switch — it controls the flow of electricity through a circuit and can be turned on or off billions of times per second. It replaced the much larger, hotter, and less reliable vacuum tube, and it made everything that followed possible. Without the transistor, there would be no **microprocessor**, no personal computer, no smartphone, and no internet.

The next breakthrough came in 1958, when engineers discovered how to place multiple transistors on a single piece of **semiconductor** material — typically silicon — creating what is known as an **integrated circuit**, or chip. Suddenly, instead of connecting individual components by hand, engineers could print entire electronic circuits onto a piece of silicon smaller than a fingernail. This dramatically reduced both the size and the cost of computers, beginning the process of miniaturisation that continues to this day.

In 1965, Gordon Moore, one of the founders of Intel, made a famous observation that became known as Moore's Law: the number of transistors that could be placed on an **integrated circuit** was doubling approximately every two years. This meant that computers would become twice as powerful every two years — not because of any single invention, but because of continuous, incremental engineering progress. For over fifty years, Moore's Law held true, driving one of the most sustained periods of technological progress in human history.

18,000 vacuum tubes in ENIAC, the first computer (1945)	2 yrs doubling time of transistors on a chip (Moore's Law)	1 bn+ transistors on a modern smartphone chip	8 bn+ people now connected to the internet worldwide
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PART THREE · THE PERSONAL COMPUTER (1970s–1990s)

In 1971, Intel released the world's first commercial **microprocessor** — the Intel 4004. For the first time, an entire **processor** was contained on a single chip of silicon. This was the key that unlocked the personal computer era. Before the microprocessor, computers were enormous and expensive — only universities, government agencies, and large companies could afford to own a **mainframe**. After it, computers became small enough and cheap enough for individuals to own.

The 1970s and 1980s saw an explosion of **innovation** as companies competed to build the best personal computer. Apple, founded by Steve Jobs and Steve Wozniak in 1976, launched the Apple II in 1977 — one of the first commercially successful personal computers. IBM followed in 1981 with its Personal Computer, which ran an **operating system** called MS-DOS, developed by a young company called Microsoft. These machines came with keyboards, screens, and — crucially — **software** that ordinary people could use without any specialist knowledge. The computer had left the laboratory and entered the home.

The development of programming languages made it possible for ordinary people — not just mathematicians — to write instructions for computers. An **algorithm** is a set of step-by-step instructions that tells a computer how to solve a problem. Programming languages like BASIC, C, and later Python allowed people to write algorithms in something closer to human language, translating them automatically into the **binary code** that the computer could actually understand.

■ HARDWARE

- The physical components of the computer — chips, circuits, screens, keyboards.
- The **processor** (CPU) that performs all calculations and runs all instructions.
- RAM — the memory that holds information the computer is currently using.
- **Data storage** devices — hard drives and solid-state drives that hold files permanently.
- Input and output devices — keyboards, mice, screens, speakers, cameras.

■ SOFTWARE

- The programs and instructions that tell the **hardware** what to do.
- The **operating system** — such as Windows, macOS, or Android — that manages everything.
- Applications — word processors, browsers, games, messaging apps.
- Algorithms — step-by-step instructions written in a **programming language**.
- **Artificial intelligence** systems that learn from data and make decisions.

PART FOUR • THE INTERNET AND THE SMARTPHONE (1990s–2010s)

In 1991, a British scientist named Tim Berners-Lee made the World Wide Web available to the public. The internet — a global network of connected computers — had existed in a limited form since the 1960s, but it was the Web that transformed it into something that ordinary people could use. For the first time, computers were not just standalone machines — they were windows onto a shared, global space where information, communication, and commerce could all take place. **Data storage** moved from individual machines onto remote servers — what we now call the cloud — making information accessible from anywhere in the world.

The next **revolution** came in 2007, when Apple launched the iPhone. It was not the first mobile phone, nor the first smartphone — but it combined a touchscreen, a powerful **processor**, internet connectivity, and a camera in a device that was genuinely easy to use. Within a decade, billions of people around the world were carrying computers in their pockets that were more powerful than the machines that had guided astronauts to the Moon. The smartphone did not just change computing — it changed communication, commerce, education, entertainment, and the way human beings relate to each other.

We are now in what many researchers believe is the most significant chapter yet in the history of computing: the age of **artificial intelligence**. AI systems — powered by vast amounts of data and enormous processing power — can now write text, generate images, translate languages, diagnose diseases, drive cars, and hold conversations that are difficult to distinguish from those with a human being. These systems do not follow a fixed **algorithm** in the traditional sense — they learn from examples, identifying patterns in data that no human programmer could have written by hand.

What makes this moment particularly remarkable is the speed of change. It took thirty years for the personal computer to become a household object. It took twenty years for the internet to reach two billion users. AI tools capable of passing university examinations and writing professional-level text reached one hundred million users in just two months. The **innovation** that began with a room-sized machine in 1945 has not slowed down — if anything, it is accelerating. The story of computers is still being written, and the most extraordinary chapters may still be ahead.

■ TALKING POINTS

Which development in the history of computers do you think was the most important? Why?

How has the smartphone changed your daily life compared to the generation before you?

Some people worry that artificial intelligence will take away jobs. Do you think this is a serious concern?

If you could go back and show someone from 1950 one piece of modern technology, what would you choose and why?